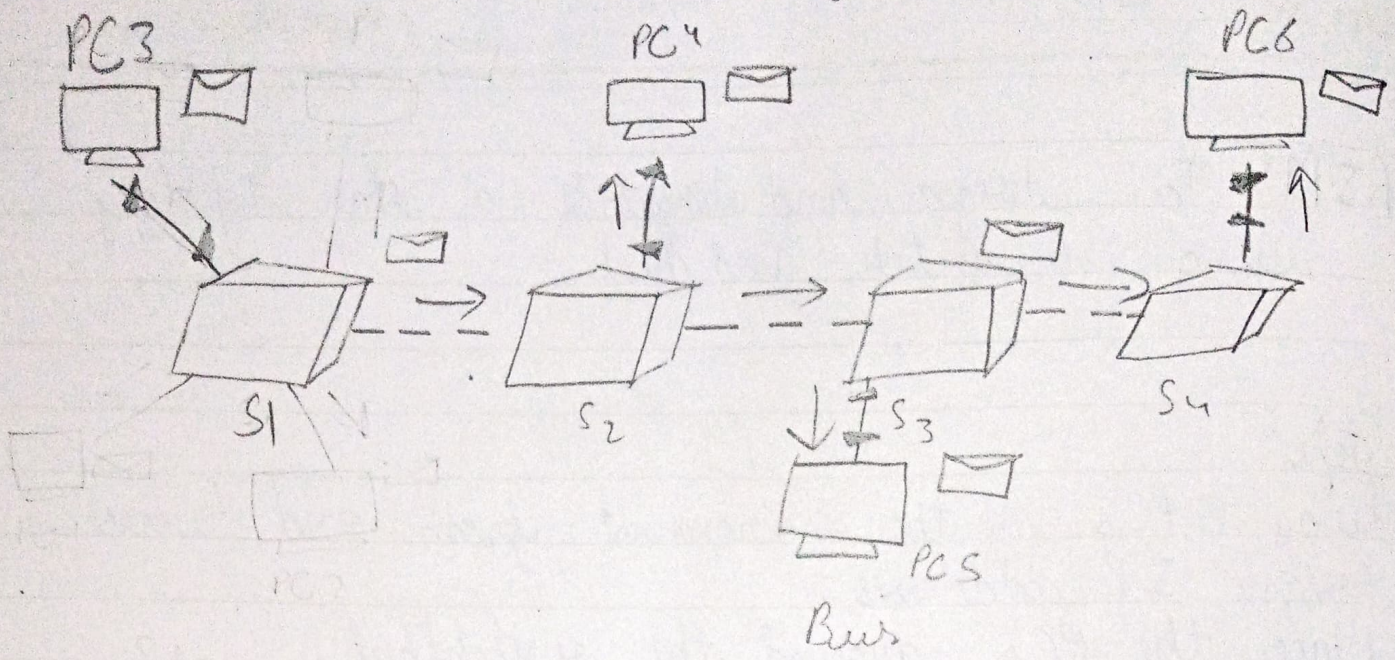
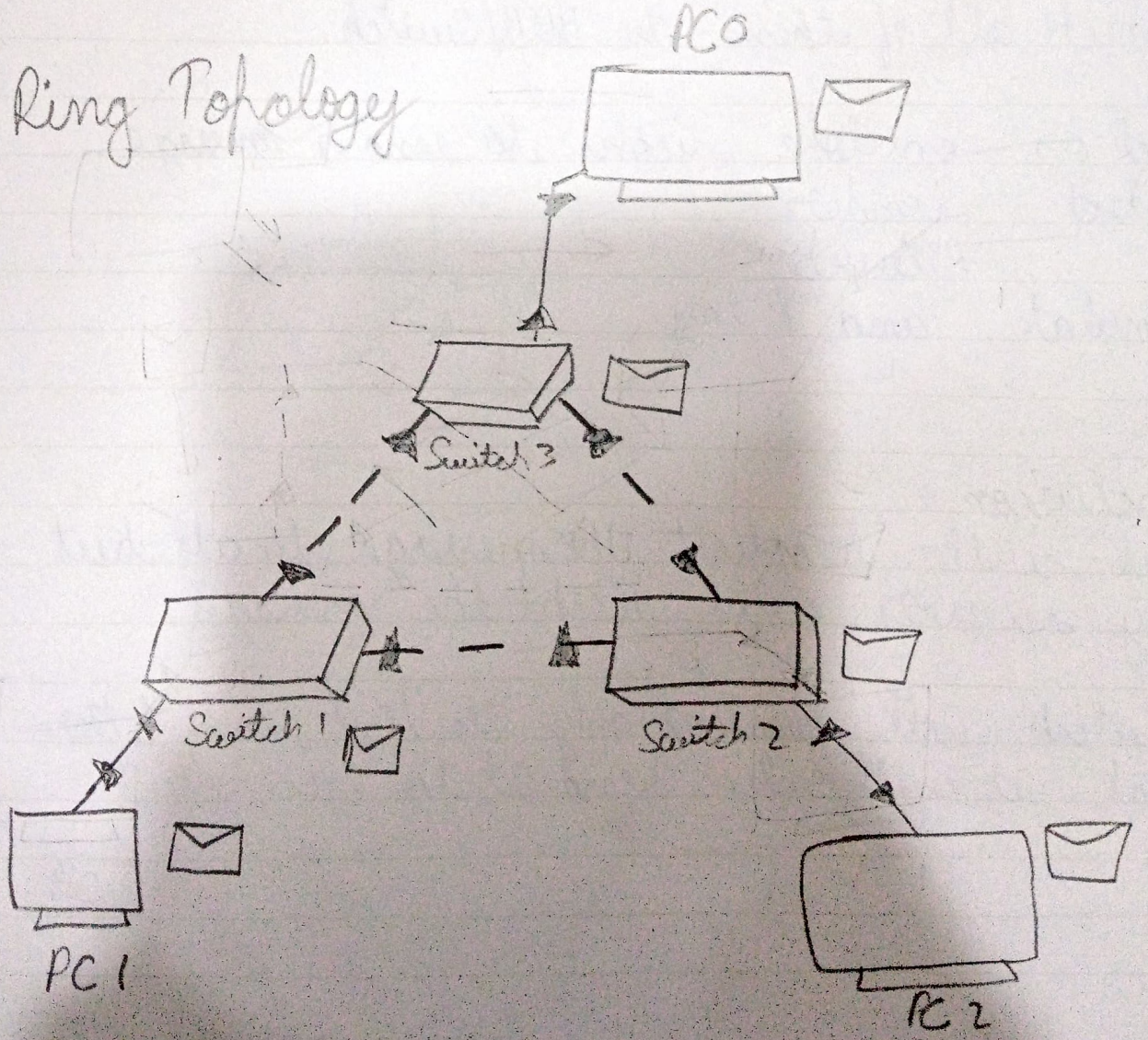
Bus Topology



Ring Topology



Date

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AIM : To simulate Bus, Ring, Mesh and Hybrid Topologies in Packet Tracer.

Bus Topology:

→ End Devices → 4 of them → Assign IPs.
→ Network devices — 4 switches
→ Place them linearly.
→ Select copper cross for switches — switch
→ Select copper-straight-through for switch-to-PC
→ Send message by envelope icon and simulate.

Observation: → Packets travel linearly.
→ All nodes receive signal
→ Delay was higher.

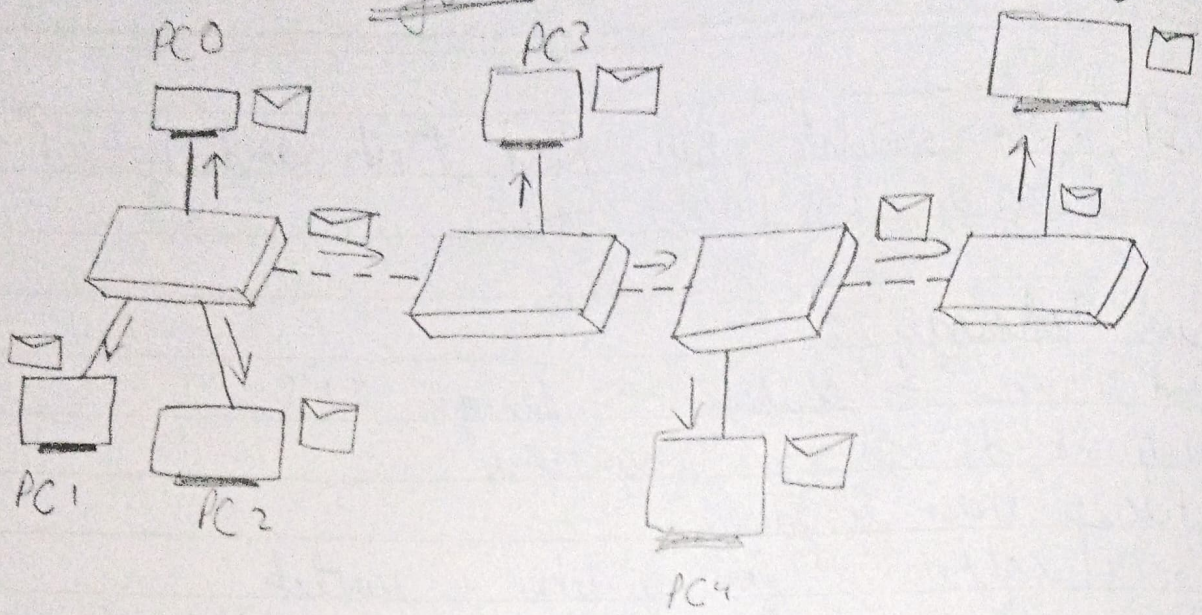
Ring Topology:

→ End devices → PCs → 3 Place 3 → Assign IPs.
→ Network device — Switch — Place 3 to 1
→ Connect switches in Ring and PC to switches, using copper cross over and structure wire respectively.
→ Send message and run simulation.

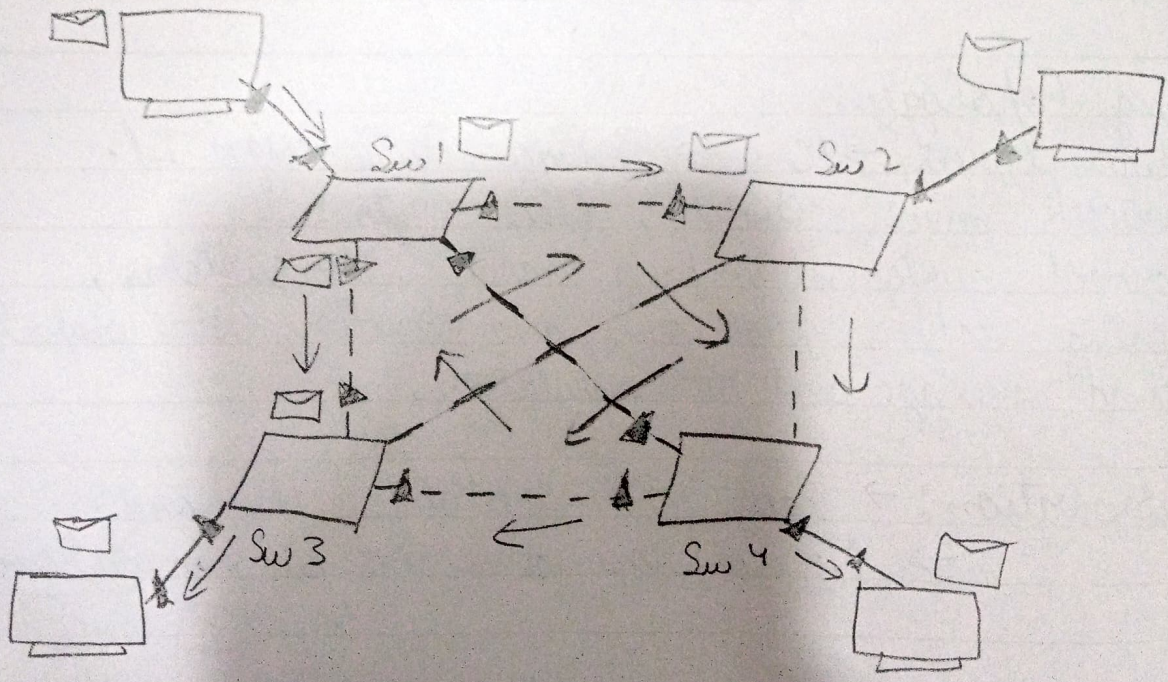
Observation: → Each node forward the frame.
→ No collision observed in simulation.

Teacher's Signature :

Hybrid



Mesh Topology



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Hybrid Topology

→ End Devices → PCs → Assign IP.
→ Network Devices → Switches
→ Instead of last PC extend its branch to fit bus topology linearly.
→ Connect the cables → use copper cross for same type of device and copper straight for different.
→ Send message and run simulation.

Observation:
→ Combination of mesh & star
→ Broadcasted to star and moved linearly in bus.

Mesh Topology

→ End Devices → PCs 5 → Assign IP
→ Network device → switch.
→ Place switch around each other and place PCs around them.
→ Connect switchs to each other including diagonally using copper cross cable.
→ Connect PCs to switches using copper straight wire.
→ Send message & run simulation.

Observation: Multiple paths are available to pass data. The packet chose shortest path.
Topology showed high reliability.

Teacher's Signature :